

Topological plasmonic chains

These are notes on the paper

Topological Plasmonic Chain with Retardation and Radiative Effects

(<https://pubs.acs.org/doi/10.1021/acsp Photonics.8b00117>) by **Giannini**. In **Section 1**, we derive **Equations 2 and 3** of this paper, which, together, define the quasistatic polarizability of a sphere including radiation losses. In **Section 2**, we calculate the coupling of individual spheres in a linear chain using the dyadic Green's function for the electric field. In **Section 3**, we derive the effective Hamiltonian of the system, writing it in terms of polylogarithms.

Section 1: Derivation of Equations 2 and 3

Our starting point is the quasistatic polarizability (including radiation losses) which was derived in the set of notes: <https://hackmd.io/@aligho/SkebhByH-g> (notes on **Andrea Alu's** paper on causality in metamaterials). In those notes, we showed that the quasistatic polarizability is given by:

$$\frac{6\pi}{k_0^3} \frac{1}{\frac{1}{(k_0 a x)^3} - i} = \frac{6\pi}{k_0^3} \frac{(k_0 a x)^3}{1 - i(k_0 a x)^3} \equiv \frac{Q}{1 - iQ(k_0^3/6\pi)},$$

which is exactly **Equation 3**. Note that in the above we used the constant, x , which is defined as (in accordance with the causality Hackmd notes):

$$x^3 = \frac{2}{3} \frac{\epsilon - 1}{\epsilon + 2} \rightarrow Q \equiv 4\pi a^3 \left[\frac{\epsilon - 1}{\epsilon + 2} \right],$$

where Q is exactly the quasistatic polarizability defined in **Equation 2**.

Section 2: The Green's function (Equations 6&7)

Now that we have the quasistatic polarizability, we wish to formulate the coupling of spheres in a linear chain. To do this, we need to know the electric field at the location of each sphere due to the dipole moments of all of the other spheres. To calculate this, we

must calculate the dyadic Green's function, $\mathbf{G}(\mathbf{r}, \omega)$, where $\mathbf{r}$ is the vector pointing from the source sphere (which creates the electric field) to the observing sphere. Once we have the Green's function, the electric field will be given by:

$$\mathbf{E}(\mathbf{r}, \omega) = i\omega\mu_0 \int \mathbf{G}(\mathbf{r} - \mathbf{r}_0, \omega) \cdot \mathbf{j}(\mathbf{r}_0, \omega) d^3\mathbf{r} \rightarrow$$

$$\mathbf{E}_n(\omega) = i\omega\mu_0(-i\omega) \sum_{m \neq n} \mathbf{G}((n - m)d\mathbf{e}_x, \omega) \cdot \mathbf{p}_m =$$

$$(\omega^2/c^2\epsilon_0) \sum_{m \neq 0} \mathbf{G}(md\mathbf{e}_x, \omega) \cdot \mathbf{p}_{n-m},$$

where $\mathbf{E}_n(\omega)$ is the electric field at lattice point n and $\mathbf{p}_m$ is the dipole moment at lattice point m . The Green's function for the electric field is given by (defining $k \equiv \omega/c$ and $r \equiv |\mathbf{r}|$):

$$\mathbf{G}_{ij}(\mathbf{r}, \omega) = \left[\mathbf{I} + \frac{1}{k^2} \nabla \nabla \right]_{ij} \frac{1}{4\pi r} e^{ikr} = \frac{1}{4\pi} \left[\frac{\delta_{ij}}{r} - \frac{1}{k^2} \partial_i \left[\frac{r_j}{r^3} - i \frac{kr_j}{r^2} \right] \right] e^{ikr} =$$

$$\frac{1}{4\pi} \left[\frac{k^{-2} [\delta_{ij} [(kr)^2 - 1 + i(kr)] + 3r_i r_j / r^2 - ik(r_i r_j / r) - (kr_i)(kr_j) - 2ik(r_i r_j / r)]}{r^3} \right] e^{ikr}$$

where the first three terms correspond to all derivatives of the type $\partial_i r_j$, the fourth term corresponds to $k^{-2} r_j e^{ikr} \partial_i (r^{-3})$, the fifth term corresponds to $k^{-2} r_j r^{-3} \partial_i e^{ikr}$, the sixth term corresponds to the derivative $ik^{-2} k r^{-2} r_j \partial_i e^{ikr}$ and the last term corresponds to $ik^{-2} k r_j e^{ikr} \partial_i r^{-2}$. We simplify this expression as follows:

$$\mathbf{G}_{ij}(\mathbf{r}, \omega) = \frac{1}{4\pi} \left[\frac{k^{-2} [\delta_{ij} [(kr)^2 - 1 + i(kr)] + 3r_i r_j / r^2 - 3ik(r_i r_j / r) - (kr_i)(kr_j)]}{r^3} \right] e^{ikr}$$

Note that if $i \neq j$, $\mathbf{G}_{ij}$ vanishes. This is obvious for the first three terms which are proportional to the Kronecker delta. For the other terms, we note that nonzero contributions are only possible if $i = j = 1$ (since r_i, r_j are zero in a linear chain otherwise).

For $i = j = 1$, we have

$$\mathbf{G}_{11}(\mathbf{r}, \omega) = \frac{1}{4\pi} \left[\frac{k^{-2}[(kr)^2 - 1 + i(kr) + 3 - 3ikr - (kr)^2]}{r^3} \right] e^{ikr} =$$

$$\frac{2}{4\pi r^3} e^{ikr} \left[1 - ikr \right] \frac{c^2}{\omega^2} \rightarrow \mathbf{E}_n \cdot \mathbf{e}_x = \frac{1}{4\pi\epsilon_0} \sum_{m \neq 0} \frac{2e^{ik|m|d}}{|m|^3 d^3} \left[1 - ik|m|d \right] (\mathbf{p}_{n-m} \cdot \mathbf{e}_x),$$

which is **Equation 6**. For $i = j = 2$, (and similarly for $i = j = 3$), we have, instead:

$$\mathbf{E}_n \cdot \mathbf{e}_y = -\frac{1}{4\pi\epsilon_0} \sum_{m \neq 0} \frac{e^{ik|m|d}}{|m|^3 |d|^3} \left[1 - ik|m|d - (k|m|d)^2 \right] (\mathbf{p}_{n-m} \cdot \mathbf{e}_y),$$

which is **Equation 7**. Applying Bloch's theorem, i.e. letting $\mathbf{p}_{(a,b),n} = e^{iqnd} \mathbf{p}_{a,b}$, where a and b are sublattice indices (note that in the above we disregarded, for simplicity, the additional sublattice indices), and noting that $\mathbf{p}_n = \alpha \mathbf{E}_n$, where α is the polarizability, yields **Equations 9 and 10**.

Section 3: Polylogarithm form of the Hamiltonian (Derivation of Equations 3 and 4 in the supplemental materials)

We discuss the polylogarithm in slightly more depth in **Appendix 1** below. For now, we just write the definition of the polylogarithm of order, s :

$$\text{Li}_s(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^s}$$

Note that this is exactly the form of the Hamiltonian with the substitutions $k = m$, $z = e^{i(k-q)d}$ for the sum $m > 0$ and $z = e^{i(k+q)d}$ for the sum $m < 0$. In particular, for the transverse directions, we see that we need three polylogarithms, of orders 3, 2, and 1. For the longitudinal Hamiltonian, we see that we only need polylogarithms of orders 3 and 2. We first tackle the longitudinal Hamiltonian:

$$\mathbf{E}_n \cdot \mathbf{e}_x = \frac{1}{4\pi\epsilon_0} \sum_{m \neq 0} \frac{2e^{ik|m|d}}{|m|^3 d^3} \left[1 - ik|m|d \right] (\mathbf{p}_{n-m} \cdot \mathbf{e}_x) \rightarrow$$

$$H^{\text{Longitudinal}} = \frac{1}{4\pi\epsilon_0} \sum_{m=1}^{\infty} \frac{2 \left[e^{i(k-q)md} + e^{i(k+q)md} \right]}{m^3 d^3} \left[1 - ikmd \right] =$$

$$\frac{2}{4\pi\epsilon_0 d^3} \left[\text{Li}_3(e^{i(k-q)d}) + \text{Li}_3(e^{i(k+q)d}) \right] - \frac{2ik}{4\pi\epsilon_0 d^2} \left[\text{Li}_2(e^{i(k-q)d}) + \text{Li}_2(e^{i(k+q)d}) \right],$$

which is exactly **Equation 3** of the **Supplemental Materials** of the paper.

We now turn to the transverse part of the Hamiltonian:

$$H^{\text{Transverse}} = -\frac{1}{4\pi\epsilon_0} \sum_{m=1}^{\infty} \frac{e^{i(k-q)md} + e^{i(k+q)md}}{m^3 d^3} \left[1 - ikmd - (kmd)^2 \right] =$$

$$-\frac{1}{4\pi\epsilon_0 d^3} \left[\left[\text{Li}_3(e^{i(k-q)d}) + \text{Li}_3(e^{i(k+q)d}) \right] - ikd \left[\text{Li}_2(e^{i(k-q)d}) + \text{Li}_2(e^{i(k+q)d}) \right] \right.$$

$$\left. - (kd)^2 \left[\text{Li}_1(e^{i(k-q)d}) + \text{Li}_1(e^{i(k+q)d}) \right] \right],$$

which is exactly **Equation 4** of the **Supplemental Materials** of the paper.

Appendix 1: The polylogarithm

The polylogarithm, $\text{Li}_s(z)$ is defined as follows:

$$\text{Li}_s(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^s}$$

An interesting fact about the polylogarithm is that it may be written in terms of the Bose integral:

$$\begin{aligned} \int_0^\infty \frac{t^{s-1}}{e^t/z - 1} dt &= \int_0^\infty \frac{t^{s-1} e^{-t} z}{1 - e^{-t} z} dt = \sum_{n=1}^\infty z^n \int_0^\infty e^{-nt} t^{s-1} dt = \\ &= \sum_{n=1}^\infty \frac{z^n}{n^s} \int_0^\infty e^{-u} u^{s-1} du = \Gamma(s) \text{Li}_s(z), \end{aligned}$$

where we used the integral representation of the Gamma function.